

Finalization Phase

DLBDSPBDM01

Build a Data Mart in SQL

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1. Introduction

This project is focused on designing and implementing a data mart for an accommodation system similar to Airbnb. The data mart is intended to store and manage all users (host, guest, admin), booking, property, payment, messaging and review data. The data mart was implemented using DuckDB as the database management system and with DBeaver as the SQL client.

The goal of the project was to design a structured database model that supports the core functionality of a booking platform and ensures consistent data management.

2. Database Design Overview

The data mart is built with data organized into multiple entities connected through foreign key relationships. The data mart contains various groups of entities representing the main functions of the booking platform.

The **user management entities** include the tables *users*, *guests*, *hosts*, and *admins*. The *users* table stores general information about all users of the system. The other tables extend this information with role-specific attributes for guests, hosts and administrators.

The **property management entities** include the tables *properties*, *property_photos*, *amenities*, *property_amenities*, and *cancellation_policy*. These tables store information about rental properties, their descriptions, locations, amenities and cancellation policies. The *property_amenities* table represents a many-to-many relationship between properties and available amenities.

The **booking related entities** include the tables *bookings*, *booking_statuses*, and *booking_cancellations*. The *bookings* table represents the central entity of the system and stores information about reservations, including references to guests, hosts and properties. The booking status is stored separately in the *booking_statuses* lookup table.

The **communication and review entities** include the tables *messages*, *property_ratings*, and *disputes*. These tables allow users to communicate regarding bookings, submit property reviews and open disputes if problems occur during a booking process.

The **payment system entities** include the tables *payments*, *refunds*, *payment_methods*, *payment_statuses*, and *payment_providers*. These tables manage

financial transactions between guests and hosts and store information about payment methods, payment providers and transaction status.

The use of separate lookup tables and relationship tables improves database normalization and allows the system to be easily extended in the future.

3. Database Functionality

The implemented data mart manages to fulfil core actions of a booking platform.

1. **User management:** The system stores user profiles and role information for guests, hosts and administrators. Each user is uniquely identified and can interact with other parts of the system depending on their role.
2. **Property listings management:** Hosts can add and manage property listings, The data mart structure allows each property to be associated with multiple amenities and images.
3. **Booking management:** Guests can create and manage their bookings for properties within a selected date range. The booking entity stores information about the guest, the host, the property and the price. Booking statuses allow the system to track the reservations.
4. **Payments:** Payment records store financial transactions related to bookings. Payments are linked to guests, hosts and bookings and include information about payment methods, providers and transaction statuses. Refunds are stored separately to manage returned payments.
5. **Communication and feedback:** Users can send messages between each other in relation to bookings and guests can submit ratings and reviews for properties following their booking completion. Administrators' role is to monitor the platform and manage disputes when necessary.

5. Summary

The final data mart structure successfully enables all use cases for the booking platform. The entity relationship model created in the conception phase was successfully translated into SQL tables and relationships during the development phase.

Overall, the implemented database provides a scalable solution for managing booking platform data and demonstrates the practical application of relational database design and SQL implementation.